

ON THE INDETERMINACY OF THE CONTINUUM HYPOTHESIS AND THE INCOMPUTABILITY OF INDETERMINACY

Can any axiom appended to ZFC genuinely resolve whether the Continuum
has a definite cardinality and can this question itself be decided?

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Abstract

If the mathematical universe is unique, resolving the Continuum Hypothesis may be understood as discovery, reaching a truth already present. If a plurality of set-theoretic universes is admitted, resolution may instead be understood as selection among models. This paper adopts a precise criterion for analysing that distinction. A closed sentence is called ZFC-invariant when all transitive set models of ZFC agree on its truth value. Assuming that a countable transitive model of ZFC exists, the formal independence of CH provides transitive models satisfying CH and its negation, so CH is not ZFC-invariant. The criterion is then turned upon itself. Every arithmetic sentence has a uniform set-theoretic translation whose truth is shared by all transitive models of ZF. From any fixed non-invariant sentence, true arithmetic many-one reduces to the invariant sentences. Their set of codes is therefore not arithmetical, and in particular is neither computably enumerable nor co-computably enumerable. The theorem concerns agreement within a specified model class, not metaphysical determinacy or truth in an intended universe.

I

What is the cardinality of the continuum? Is there a cardinal strictly between $\aleph_0$ and $|\mathcal{P}(\omega)| = 2^{\aleph_0}$?* The conjecture that there is not, that $2^{\aleph_0} = \aleph_1$, is the Continuum Hypothesis. If ZFC is a foundation for mathematics, it is natural to ask why ZFC can neither prove nor refute it.

What can an axiom system determine about the universe it describes? An axiom constrains models, and every structure satisfying all the axioms is a model of the theory. If a first-order theory is syntactically consistent, then by the Completeness Theorem it has a model.[†] The Löwenheim–Skolem theorem further shows that a first-order theory with an infinite model has models in different infinite cardinalities.[‡] These general model-theoretic facts do not prove the independence of CH. Gödel and Cohen supplied the separate relative consistency results

$$\text{Con}(\text{ZFC}) \implies \text{Con}(\text{ZFC} + \text{CH})$$

and

$$\text{Con}(\text{ZFC}) \implies \text{Con}(\text{ZFC} + \neg\text{CH}).$$

They do not by themselves assert the existence of transitive set models. The stronger model-theoretic formulation used below therefore carries an explicit metatheoretic assumption.

Assume that a countable transitive model of ZFC exists. Gödel’s constructible hierarchy, formed internally to such a model, gives a transitive set model in which CH holds.[§] A suitable forcing extension adds enough subsets of ω while preserving the relevant cardinals and gives a transitive set model in which CH fails. Adding one Cohen real alone need not make CH false.[¶] The proof-theoretic conclusion remains

$$\text{ZFC} \not\vdash \text{CH}, \quad \text{ZFC} \not\vdash \neg\text{CH}.$$

The independence also creates a philosophical choice. If the mathematical universe is a single pre-given totality, as Gödel maintained, then CH may have a determinate truth value in the intended universe V , and independence records the weakness of the present axioms. Resolution would then be discovery. If the plurality produced by forcing and inner models is treated as mathematically substantive, resolution may instead consist in selecting a class of universes. The mathematics does not decide between these positions. The paper compares them using transitive-model invariance.

A set M is transitive when $x \in M$ and $y \in x$ imply $y \in M$. A transitive set model uses the ambient membership relation restricted to its domain. This makes it possible to compare finite mathematics across models, but it does not establish that transitive models are the only legitimate interpretations of set theory.^{||} A closed sentence is called ZFC-invariant when every two transitive set models of ZFC agree on its truth value. Invariance is a model-theoretic property. Identifying it with genuine determinacy is an additional philosophical commitment.

Criterion. For the purposes of this paper, an axiom A *genuinely resolves* CH when $\text{ZFC} + A$ proves CH or $\neg\text{CH}$ and at least one transitive set model satisfies $\text{ZFC} + A$, while the established truth value already holds in every transitive set model of ZFC.

The criterion has an immediate precedent. PA cannot prove its own consistency, yet ZFC can.** The sentence $\text{Con}(\text{PA})$ asserts that no natural number encodes a PA derivation of a contradiction. ZFC constructs the standard arithmetic structure on ω , verifies the axioms of PA, and formalises soundness for first-order derivations. It therefore proves $\text{Con}(\text{PA})$. Every transitive model of ZF has the same ω and agrees on the arithmetic operations, so all such models agree on this arithmetic sentence. This is

*G. Cantor, “Ein Beitrag zur Mannigfaltigkeitslehre,” *J. reine angew. Math.* 84 (1878), 242–258.

[†]K. Gödel, “Die Vollständigkeit der Axiome des logischen Funktionenkalküls,” *Monatshefte für Mathematik und Physik* 37 (1930), 349–360.

[‡]T. Skolem, “Einige Bemerkungen zur axiomatischen Begründung der Mengenlehre,” *Matematikerkongressen i Helsingfors* 4 (1922), 217–232.

[§]K. Gödel, “Consistency-proof for the generalized continuum-hypothesis,” *Proc. Nat. Acad. Sci.* 25 (1939), 220–224; *The Consistency of the Axiom of Choice and of the Generalized Continuum-Hypothesis with the Axioms of Set Theory* (Princeton, 1940).

[¶]P. Cohen, *Proc. Nat. Acad. Sci.* 50 (1963), 1143–1148; *Set Theory and the Continuum Hypothesis* (W. A. Benjamin, 1966).

^{||}The Mostowski collapse theorem applies to externally well-founded, extensional, set-like relations. The existence of a transitive set model of ZFC is stronger than $\text{Con}(\text{ZFC})$ and is used here as an explicit metatheoretic premise.

**K. Gödel, “Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I,” *Monatshefte für Mathematik und Physik* 38 (1931), 173–198.

the contrast that motivates the criterion. Arithmetic truth is stable across the model class, while CH is not.

Does $\mathcal{P}(\omega)$ admit invariance over the original class? A subset of ω lies above ω in the cumulative hierarchy, and transitive models can contain different collections of such subsets. Under the explicit countable-transitive-model premise, there are transitive ZFC models satisfying CH and others satisfying its negation. CH therefore fails the stated criterion. Any metaphysical conclusion depends on accepting that criterion.

II

The constructible universe gives the canonical inner-model route to CH. At each successor stage of the hierarchy it admits the subsets that are first-order definable over the preceding stage using parameters already present. At a limit stage it takes the union of the earlier levels.*

$$L_0 = \emptyset, \quad L_{\alpha+1} = \text{Def}(L_\alpha), \quad L_\lambda = \bigcup_{\alpha < \lambda} L_\alpha, \quad L = \bigcup_{\alpha \in \text{Ord}} L_\alpha.$$

The construction is cumulative, and each level depends only on earlier levels. It is important not to confuse constructibility with parameter-free definability. The parameters in $\text{Def}(L_\alpha)$ allow many sets that cannot be individually named without parameters. The assertion $V = L$ says that every set appears somewhere in this iterated hierarchy.

The condensation argument explains the consequence for the continuum. If $x \subseteq \omega$ and $x \in L$, then x appears in some L_α with $\alpha < \omega_1^L$. Internally to L , each such level is countable, and there are $\aleph_1^L$ many relevant stages.[†] Hence

$$L \models |\mathcal{P}(\omega)| \leq \aleph_1.$$

Cantor's theorem gives the reverse inequality, and therefore

$$L \models 2^{\aleph_0} = \aleph_1.$$

The ambient L is a proper-class inner model, not a member of the class of transitive set models used in this paper. Under the countable-transitive-model premise, the corresponding hierarchy L^{M_0} formed inside the ground model M_0 supplies the required transitive set model of ZFC + CH.

The forcing direction changes the collection of reals. If M is a transitive model of the Power Set axiom and $a \in M$, then externally

$$\mathcal{P}^M(a) = \mathcal{P}(a) \cap M.$$

The equality does not mean that M fails its own Power Set axiom. It means that the quantifiers of M range over its own domain. A generic extension $M[G]$ can contain subsets of ω that were absent from M , while both models satisfy the same axiom. Suitable forcing can add enough reals and preserve the relevant cardinals so that $M[G] \models \neg\text{CH}$. The existence of one nonconstructible real is not by itself sufficient.

The constructible restriction has consequences beyond CH. Scott proved

$$V = L \implies \text{there is no measurable cardinal.}$$

[‡] Relative to the criterion of §I, $V = L$ selects a mathematically consequential, nonempty subclass of the transitive ZFC models lying on the CH side of a prior disagreement, rather than establishing a truth shared by the original class.[§]

III

*K. Gödel, "Consistency-proof for the generalized continuum-hypothesis," *Proc. Nat. Acad. Sci.* 25 (1939), 220–224; *The Consistency of the Axiom of Choice and of the Generalized Continuum-Hypothesis with the Axioms of Set Theory* (Princeton, 1940), Ch. II.

[†]The superscript matters. The ordinal ω_1^L need not equal the first uncountable ordinal of the ambient universe.

[‡]D. Scott, "Measurable cardinals and constructible sets," *Bull. Acad. Polon. Sci.* 9 (1961), 521–524.

[§]P. Maddy, "Believing the axioms I & II," *J. Symbolic Logic* 53 (1988), 481–511 and 736–764.

If restriction narrows too much, perhaps enrichment will succeed. Large cardinal axioms form a family of principles asserting strong infinitary properties. A strongly inaccessible cardinal κ satisfies $V_\kappa \models \text{ZFC}$, and many familiar stronger large cardinal notions imply inaccessibility. Their consistency strength can extend far beyond ZFC, but strength alone does not determine the value of the continuum.

Forcing explains the separation. A forcing notion $\mathbb{P}$ can add new subsets of ω while leaving a large cardinal above it intact. Lévy and Solovay established a precise instance of this phenomenon.*

Lévy–Solovay (1967). Let κ be a measurable cardinal and let $\mathbb{P}$ be a forcing notion with $|\mathbb{P}| < \kappa$. Then forcing with $\mathbb{P}$ preserves the measurability of κ .

In a suitable ground model, forcing that changes CH can be chosen small relative to the measurable cardinal. The associated elementary embedding can be lifted, producing a measure in the extension. Thus the measurable cardinal can coexist with CH in one model and with its negation in another. This separates measurability from the value of CH. The result is specific to measurable cardinals.

Forcing axioms take a different route. Martin’s Maximum asserts a strong filter-existence principle for stationary-set-preserving partial orders and implies

$$2^{\aleph_0} = \aleph_2.$$

Foreman, Magidor, and Shelah established its consistency relative to a supercompact cardinal.† The axiom therefore gives a mathematically rich model class in which CH is false. It does not make CH invariant over the larger class of all transitive ZFC models.

Every transitive model M of ZF has the standard natural numbers.

Absoluteness of ω . For every transitive model $M \models \text{ZF}$,

$$\omega^M = \omega.$$

Every standard finite ordinal is produced from 0 by finitely many successor operations and belongs to M . Transitivity and the internal definition of ω prevent additional nonstandard finite ordinals from appearing. Addition, multiplication, ordering, and all primitive recursive relations on ω consequently agree between transitive models. An induction on the structure of formulas shows that every closed first-order arithmetic sentence has the same truth value in all of them.

The power set does vary. The quantifier over all subsets of ω ranges beyond ω itself, and different transitive models may contain different subsets. Under the explicit countable-transitive-model premise, the formal independence construction gives

$$\exists M, N \in \text{TrZFC} (M \models \text{CH} \wedge N \models \neg \text{CH}).$$

CH is therefore not ZFC-invariant. Arithmetic sentences are stable across the chosen transitive models because their quantifiers range over the shared standard ω . CH is not stable because it quantifies over sets whose availability varies with the model. Feferman’s discussion of CH as an indefinite problem and Hamkins’s multiverse view give different philosophical interpretations of this contrast.‡

IV

The Gödel and Cohen constructions exhibit transitive models of ZFC that disagree on CH. Their membership relations are restrictions of the ambient relation $\in$, and both contain the standard ω . This is the exact common ground needed for the comparison. The models need not have the same domain, the same higher set structure, or the same large cardinals, and a forcing extension does not differ from its ground model only in $\mathcal{P}(\omega)$. What the constructions establish is narrower and decisive. ZFC permits transitive models with different collections of subsets of ω , and the resulting models can disagree on the cardinality of that collection.

Any additional axiom that settles CH restricts the original model class. If A is consistent with the stated transitive-model premise, the passage is from the class TrZFC to the nonempty subclass

*A. Lévy and R. Solovay, “Measurable cardinals and the continuum hypothesis,” *Israel J. Math.* 5 (1967), 234–248.

†M. Foreman, M. Magidor, and S. Shelah, “Martin’s Maximum, saturated ideals, and nonregular ultrafilters,” *Annals of Mathematics* 127 (1988), 1–47.

‡S. Feferman, “Is the Continuum Hypothesis a definite mathematical problem?” *Exploring the Frontiers of Incompleteness*, Harvard University, 2011. J. D. Hamkins, “The set-theoretic multiverse,” *Rev. Symbolic Logic* 5 (2012), 416–449.

$$\text{TrZFC}[A] = \{M \in \text{TrZFC} \mid M \models A\}.$$

When all models in this subclass satisfy one side of CH, A selects that side relative to the original class. This description is independent of whether A is motivated by constructibility, forcing axioms, large cardinals, or an inner-model programme. It records the formal effect of the axiom. It does not yet decide whether the selected subclass is the correct universe of sets.

The Platonist therefore objects that the comparison concedes too much. The models need not be equally intended. One of them, or neither of them, may be the true universe V , and CH may possess a unique truth value there even when other models satisfy the opposite value. On this view, independence is an epistemic limitation of ZFC rather than an indeterminacy in mathematical reality. Gödel held a position of this kind, and the existence of competing models does not refute it.*

The objection immediately raises the question that motivated the original argument. What singles out V as intended rather than a transitive model in which CH has the other value? The formal independence proof supplies no answer. The shared standard arithmetic cannot distinguish the models, while their collections of reals can. Yet it would be too strong to conclude that specifying $\mathcal{P}(\omega)$ specifies the entire universe. Forcing can alter higher parts of the cumulative hierarchy without adding reals, and different universes may agree on their reals while differing elsewhere. The correct conclusion is that any identification of the intended V requires principles beyond the bare fact that its membership relation is genuine.

If the realist appeals to mathematical intuition, intrinsic justification, or an intended conception of set, the appeal lies outside the invariance criterion and cannot make agreement across all transitive ZFC models a consequence of the independence theorem. Universe and multiverse theorists may therefore share the mathematics while differing in interpretation.

The contrast with ω remains instructive. Every transitive model of ZF has the same natural numbers and the same first-order arithmetic structure. Its arithmetic sentences consequently have one truth value throughout the model class. The collection $\mathcal{P}(\omega)$ is not stable in this way. This contrast supports the paper's criterion, but it does not prove that the criterion is the only possible account of mathematical determinacy. Full second-order semantics may express categorical principles, but its range of second-order quantification already depends on a background collection of sets. It therefore relocates rather than automatically resolves the present question.

Discovery and selection are relative to the chosen model class. Under the criterion of § I, a CH-deciding axiom is selection over TrZFC , but it may be discovery over a smaller class accepted for independent reasons or over a separately justified unique intended V .

V

The preceding sections used model-invariance as a test for the distinction between discovery and selection. CH was examined by means of specific inner-model and forcing constructions. A further question now arises. Can the same test be applied uniformly? Given the code of an arbitrary closed sentence of set theory, can an algorithm determine whether all transitive models of ZFC agree on its truth value?

The question can be stated independently of CH. The argument is made externally, in a metatheory in which classes of set models and their satisfaction relations can be discussed. Let $\mathcal{K}$ be such a class of transitive set models of ZFC. When $\mathcal{K}$ is proper, the notation below denotes the externally defined collection of natural-number codes. For a closed sentence φ , write

$$\text{Inv}_{\mathcal{K}}(\varphi) \quad \text{when} \quad \forall M, N \in \mathcal{K} \ (M \models \varphi \longleftrightarrow N \models \varphi).$$

Fix an effective Gödel numbering of the formulas of set theory and let

$$I_{\mathcal{K}} = \{\ulcorner \varphi \urcorner \mid \varphi \text{ is closed and } \text{Inv}_{\mathcal{K}}(\varphi)\}.$$

The computational problem is the problem of recognising membership in $I_{\mathcal{K}}$.

Every first-order arithmetic sentence θ has an effective translation θ^{ω} into the language of set theory. Its number quantifiers are restricted to ω , and addition, multiplication, and order receive their usual set-theoretic definitions. The map from $\ulcorner \theta \urcorner$ to $\ulcorner \theta^{\omega} \urcorner$ is computable. Since every transitive model of ZF has the standard ω and the standard arithmetic operations, for every $M \in \mathcal{K}$,

*K. Gödel, "What is Cantor's continuum problem?" *American Mathematical Monthly* 54 (1947), 515–525.

$$M \models \theta^\omega \iff \mathbb{N} \models \theta.$$

Thus the full truth of first-order arithmetic, not merely one syntactic fragment of it, is shared by all members of $\mathcal{K}$.

Suppose now that $\mathcal{K}$ contains two models that disagree on a fixed closed sentence σ . Equivalently, $\neg \text{Inv}_{\mathcal{K}}(\sigma)$. Let $\text{Th}(\mathbb{N})$ be the set of Gödel numbers of closed arithmetic sentences true in the standard natural numbers. Define a total computable function F as follows. If n is the code of a closed arithmetic sentence θ , let

$$F(n) = \ulcorner \theta^\omega \vee \sigma \urcorner.$$

If n is not such a code, let $F(n) = \ulcorner \sigma \urcorner$. The syntactic tests and all operations on formula codes used here are primitive recursive.

Theorem. If a class $\mathcal{K}$ of transitive models of ZFC contains disagreement on some closed sentence, then

$$\text{Th}(\mathbb{N}) \leq_m I_{\mathcal{K}}.$$

Consequently, $I_{\mathcal{K}}$ is not arithmetical. In particular, it is neither computably enumerable nor co-computably enumerable.

Proof. Let n code a closed arithmetic sentence θ . If $\mathbb{N} \models \theta$, then every member of $\mathcal{K}$ satisfies θ^ω . Every member therefore satisfies $\theta^\omega \vee \sigma$, so $F(n) \in I_{\mathcal{K}}$. If $\mathbb{N} \not\models \theta$, then every member of $\mathcal{K}$ falsifies θ^ω . In each such model the sentence $\theta^\omega \vee \sigma$ has exactly the truth value of σ . Since σ varies across $\mathcal{K}$, $F(n) \notin I_{\mathcal{K}}$. Invalid codes belong neither to $\text{Th}(\mathbb{N})$ nor, by the second clause in the definition of F , to the inverse image of $I_{\mathcal{K}}$. Hence

$$n \in \text{Th}(\mathbb{N}) \iff F(n) \in I_{\mathcal{K}}.$$

The set $\text{Th}(\mathbb{N})$ is not arithmetical by Tarski's undefinability theorem. The inverse image of an arithmetical set under a total computable function is arithmetical. Therefore $I_{\mathcal{K}}$ cannot be arithmetical.* $\square$

The halting problem gives the original construction in a more concrete form. Fix a total effective enumeration $(M_e)_{e \in \omega}$ of Turing machines, so that every natural number denotes a machine, together with an effective coding of their finite configurations. There is an arithmetic formula $C(x, s)$ representing the primitive recursive relation which holds when s codes a finite legal computation of the machine with index x on empty input and its last configuration is final. For each meta-level index e , let $\bar{e}$ be its arithmetic numeral and let H_e be the closed arithmetic sentence

$$H_e = \exists s C(\bar{e}, s).$$

The machine with code e halts on empty input exactly when $\mathbb{N} \models H_e$. Since H_e is arithmetic, every model in $\mathcal{K}$ agrees on its translated set-theoretic form. The two computable maps

$$f(e) = \ulcorner H_e^\omega \vee \sigma \urcorner, \quad g(e) = \ulcorner (\neg H_e)^\omega \vee \sigma \urcorner$$

satisfy

$$e \in K_0 \iff f(e) \in I_{\mathcal{K}}, \quad e \notin K_0 \iff g(e) \in I_{\mathcal{K}},$$

where K_0 is the ordinary empty-input halting set. The first equivalence shows directly that no decision procedure for invariance can exist. Taken together, the two equivalences also show that $I_{\mathcal{K}}$ is neither computably enumerable nor co-computably enumerable. If $I_{\mathcal{K}}$ were co-computably enumerable, its computable preimage under f would make K_0 co-computably enumerable. If $I_{\mathcal{K}}$ were computably enumerable, its computable preimage under g would make the complement of K_0 computably enumerable. Both consequences contradict the halting theorem.[†]

*A. Tarski, "The concept of truth in formalized languages," in *Logic, Semantics, Metamathematics* (Oxford, 1956), 152–278.

[†]The historical source is A. M. Turing, "On computable numbers, with an application to the Entscheidungsproblem," *Proc. London Math. Soc.* 42 (1936–1937), 230–265. The modern empty-input formulation is standard in H. Rogers, *Theory of Recursive Functions and Effective Computability* (McGraw–Hill, 1967).

CH supplies a canonical instance of σ , but the theorem itself is not about CH. Assume, as in the earlier sections, that a countable transitive model of ZFC exists. The constructible inner-model argument and a suitable forcing extension then provide transitive models of ZFC that disagree on CH. Taking $\mathcal{K}$ to be the class of transitive set models of ZFC and σ to be CH yields

$$\text{Th}(\mathbb{N}) \leq_m I_{\text{TrZFC}}.$$

The reduction therefore isolates the exact division of labour. Arithmetic provides sentences whose truth is fixed across the model class. One fixed non-invariant sentence provides the variation. Disjunction converts the truth of the arithmetic sentence into the invariance of a set-theoretic sentence. No special property of CH beyond its non-invariance is used.

For CH the disagreement is known by independent mathematical constructions. Relative to the criterion of §I, no algorithm recognises discovery uniformly across all sentence codes. The transitive-model-invariance index set is not arithmetical.

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